



High-temperature Si-Pr eutectic filler for brazing RBSiC composites: Wettability and bonding mechanisms



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Abstract

This study investigates the brazing of Si-Pr eutectic filler with Reaction-bonded Silicon Carbide (RBSiC), offering a comprehensive analysis of the wetting mechanism, microstructural evolution, mechanical properties, and fracture behavior of the joints. The results demonstrate that the Si-Pr eutectic filler achieves exceptional wettability with RBSiC, with no new phases formed at the interface, indicating stable and reliable interfacial interactions. Notably, the joint shear strength reaches 62.6 MPa at 1290 °C, with fracture occurring within the substrate, highlighting the superior mechanical properties of the brazed joints. The bonding mechanism involves Si adsorption and atomic bond formation facilitated by the diffusion of the filler and interface reconstruction. This work not only enhances our understanding of Si-Pr eutectic brazing alloys but also provides valuable insights

for advancing composite material joining technologies, paving the way for more robust and efficient material connections in high-performance applications.



Keywords

Reaction-sintered silicon; Si-Pr eutectic filler; Microstructure; Mechanical properties

1. introduction

Reaction-bonded Silicon carbide is a high-performance Si/SiC two-phase composite ceramic formed by reaction synthesis [1], [2], [3]. It is widely used in many demanding applications due to its exceptional physical, chemical and mechanical properties, including high wear resistance, high temperature resistance, excellent chemical stability and high thermal conductivity [4], [5], [6], [7], which make RBSiC an ideal choice for many advanced engineering applications, including aerospace, automotive, electronics, energy and industrial processes [8], [9], [10], [11], [12]. However, due to the challenging machinability of RBSiC, there are limitations in the fabrication of large and complex shaped components, joining techniques for RBSiC have emerged as an effective solution to address these challenges [13], [14]. SiC welding has been mainly involved in direct reaction joining [15], [16], combustion reaction joining [17], pre-ceramic polymer joining [18], brazing [19], and diffusion joining [20], [21]. Brazing stands out due to its lower processing temperatures and reduced thermal damage to the material compared to reactive joining techniques. Moreover, it offers superior energy efficiency and greater control over the process than combustion reaction joining. Additionally, brazing is faster, more economical, and safer than diffusion joining and pre-ceramic polymer joining, establishing it as a preferred method for joining ceramics. Its simplicity and versatility make brazing an ideal choice for most ceramic self-joining applications. However, a key challenge in brazing SiC lies in the control of interfacial reactions [22], which significantly influence the mechanical performance of the joints.

Ceramic materials typically exhibit poor wettability with metal brazing fillers due to their chemical stability. Brazing filler commonly used in ceramic brazing contain reactive elements (Ti, Zr, V, etc.), which significantly improve the wettability and interfacial reactions between metal brazing fillers and ceramics. However, the reactive elements are prone to react violently with SiC, forming brittle phases such as carbides and silicides [23], [24]. In addition, the differing thermal expansion coefficients between the metal brazing filler and the RBSiC substrate cause thermal stress mismatches during heating and cooling, which concentrate in the intercalation layer of the brazed joints, leading to performance degradation.

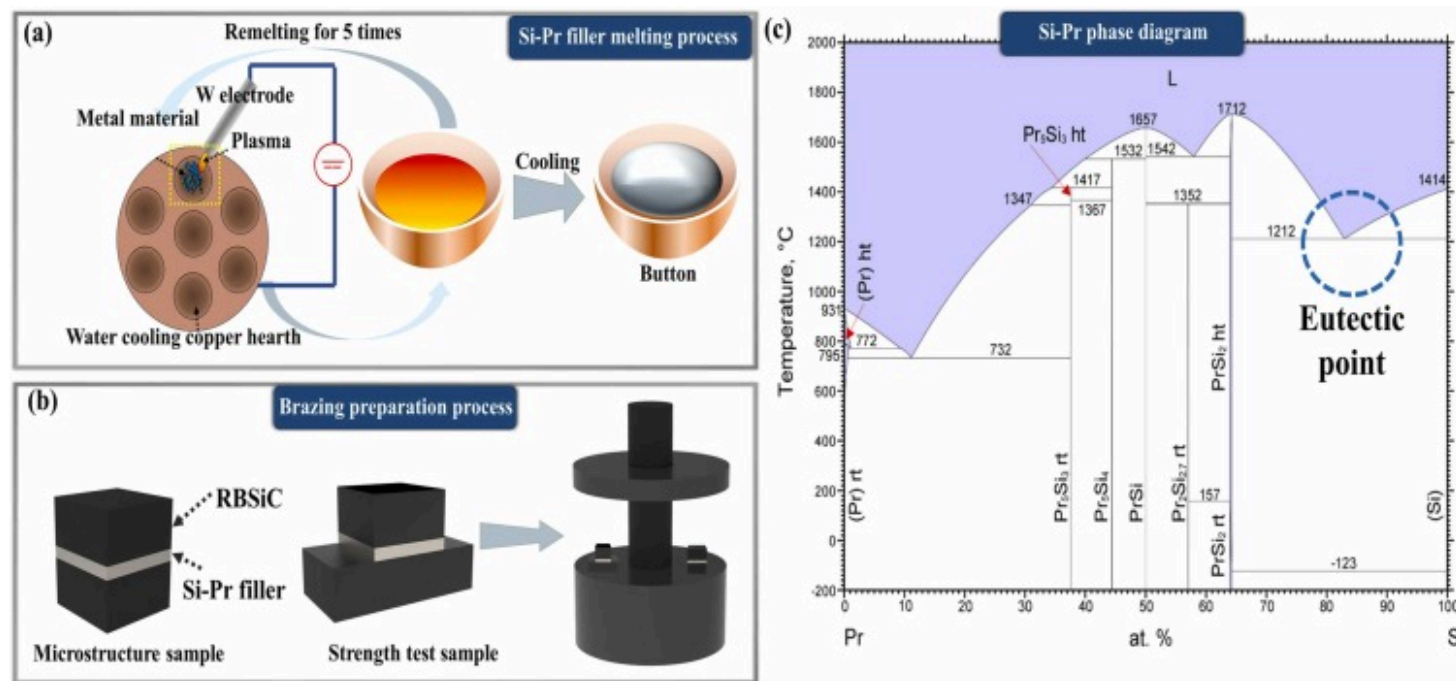
In order to overcome this challenge, relevant literature indicates that the addition of a controlled amount of Si can effectively mitigate the reactivity of metal elements with SiC while simultaneously enhancing the wettability of the brazing filler. Furthermore, Si incorporation contributes to the alignment of thermal expansion coefficients between the brazing filler and SiC, which is advantageous for reducing thermal stresses. Some studies have explored non-reactive joining techniques for SiC using binary alloys with high Si content as brazing fillers. This approach is attributed to the superior non-reactive wettability exhibited by these fillers in contact with SiC. Indeed, a considerable body of research has demonstrated that a variety of binary Si-based alloys, such as Ni-Si [25], Co-Si [26] Fe-Si [27], Au-Si [28], Ag-Si [29], and Ti-Si [30], show promising non-reactive wetting behavior with SiC substrates. It has been established that the incorporation of Si not only impedes the formation of brittle phases like graphite but also enhances the bond integrity through high adhesion energy between the brazing filler and the substrate. The diffusion kinetics governing the joint formation are primarily influenced by the dynamics of oxide layer removal at the interface [31]. While significant advancements have been made in the use of Si-based alloys for SiC brazing, there is a notable gap in research concerning the use of Si-Pr filler for RBSiC joining. The Si-Pr eutectic system, with its eutectic point at 1212 °C, offers a unique advantage in forming high-temperature-resistant joints, making it a promising candidate for the development of more reliable brazed joints for SiC-based materials.

This study explores the innovative use of Si-Pr filler for brazing RBSiC, offering a comprehensive evaluation of its wettability to assess the potential for forming robust, reliable joints. A detailed investigation was conducted into the effect of brazing temperature on the evolution of interfacial microstructure, phase composition, and the resulting shear strength of the joints. Additionally, the underlying mechanisms of joint formation were examined to uncover the nature of interfacial bonding. The findings provide valuable insights into optimizing brazing processes.

2. Materials and methods

2.1. Samples preparation

According to the phase diagram ratio shown in Fig. 1(c), the Si-17Pr (at%) filler was prepared by arc melting. Pr blocks (>99% purity, Zhongjin Yan Technology Co., Ltd.) and Si blocks (>99% purity, Zhongjin Yan Technology Co., Ltd.) were raw materials. Fig. 1(a) shows the schematic diagram of the brazing alloy melting process. The alloy was melted in a water-cooled copper crucible under the following conditions: a current of 200 A, a voltage of 30 V, and a melting time of 15 s. The alloy was melted in a vacuum melting furnace at 1500 °C with a vacuum pressure of 5×10^{-3} Pa. To ensure the homogeneity of the alloy, the samples were remelted 5 times, with the samples being fully turned over during each melting process. After completing the melting, spherical ingots were obtained. RBSiC is formed by infiltrating molten silicon into a porous C/SiC green body, where Si reacts with C to generate SiC, resulting in a Si/SiC two-phase composite material [3]. The RBSiC material used in this study was supplied by China Sanze New Materials Co., Ltd., with a surface roughness of $R_a=0.8$. The material was cut into blocks of 4 mm × 4 mm × 3 mm and 4 mm × 8 mm × 3 mm using diamond wire cutting. The alloy ingots were also cut into 4 mm × 4 mm × 0.5 mm brazing flakes using diamond wire cutting. Both the substrate and brazing flakes were polished with sandpaper, followed by ultrasonic cleaning in acetone for 10 min. The cleaned samples were then blown dry and set aside for further use. The prepared masterbatch and brazing flakes were assembled in a sandwich configuration: ceramic/brazing alloy/ceramic, and the entire assembly was placed into a graphite mold (Fig. 1(b)). The assembly was then placed into a vacuum furnace for brazing. The brazing process was carried out at a vacuum level of 2.7×10^{-5} Pa, with the brazing temperature set between 1250 °C and 1350 °C and the brazing time set to 15 min. During the heating and cooling process, the temperature rise/cooling rate was set to 10 °C/min below 800 °C and 5 °C/min above 800 °C.



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Fig. 1. Schematic of the joining process. (a) Schematic of brazing alloy melting; (b) Schematic of brazing preparation; (c) Si-Pr phase diagram [32].

2.2. Microstructure characterization and mechanical properties testing

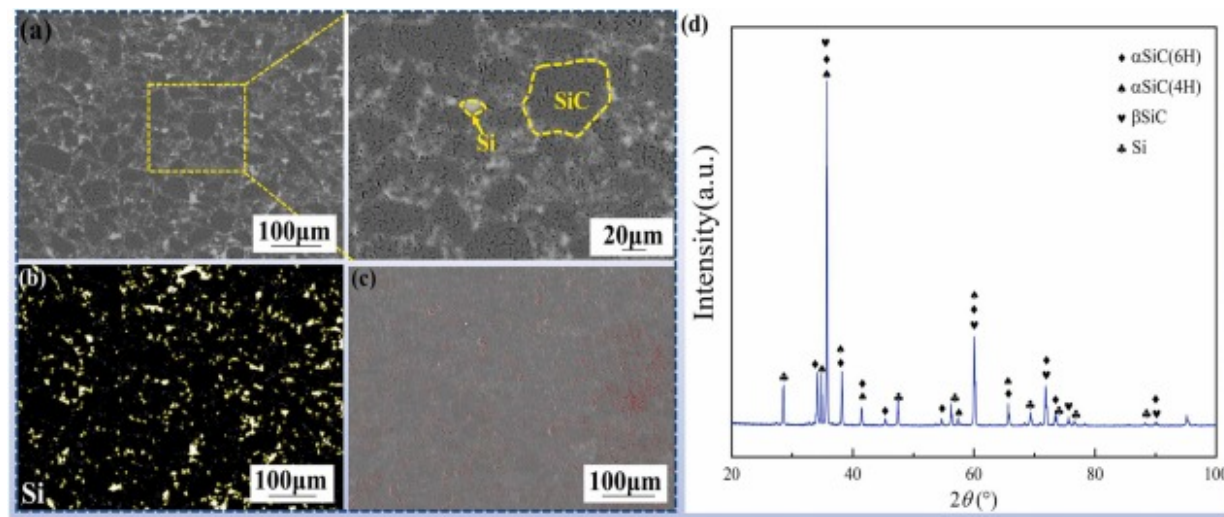
The microstructure of the filler and fracture surface were observed using scanning electron microscope (SEM, Tescan amber), and elemental and phase analysis was conducted using an energy-dispersive X-ray spectroscope (EDS). Elemental mapping in the joints was analyzed using an electron probe microanalyzer (EPMA, JXA-8230). The melting point of the eutectic alloy was determined via differential scanning calorimetry (DSC, Netzsch STA449F3), utilizing a heating rate of 10 °C/min. The phase composition analysis was carried out using an Empyrean X-ray diffractometer (XRD, Empyrean). For detailed microstructural analysis, transmission electron microscopy (TEM, Talos F200X) was employed, with specimen preparation conducted utilizing the focused ion beam (FIB, Tescan amber) technique. Finally, the shear strength of the brazed joints was tested using a universal

testing machine (AGX-plus) with a maximum load capacity of 20 kN and a loading rate of 0.5 mm/min, The shear strength was measured by dividing the maximum force by the bonded area. For each testing condition, five samples were evaluated, and the mean value along with the standard deviation was reported. Error bars indicate the variability of the results.

3. Results and discussion

3.1. Characterization of substrate and Si-Pr filler

Fig. 2(a) shows backscattered electron (BSE) images of RBSiC, indicating that it primarily consists of SiC grains surrounded by free Si. Fig. 2(b) presents the image analysis performed using ImageJ software to quantify the free Si content in Fig. 2(a), estimating a fraction of 4.34 vol%. Additionally, secondary electron (SEM) imaging of the substrate surface estimates the porosity at 2.67 vol% (Fig. 2(c)). XRD analysis (Fig. 2(d)) further confirms that the SiC phase predominantly consists of β -SiC with a cubic structure and α -SiC with a hexagonal structure.

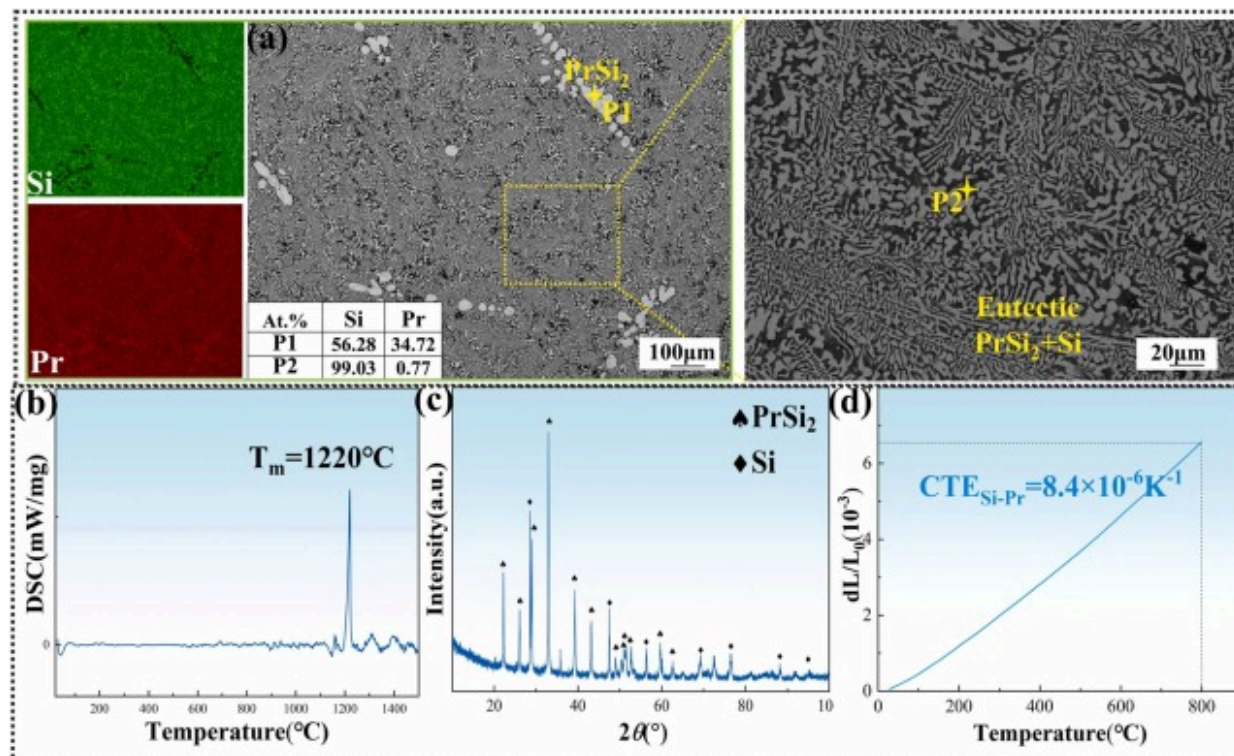


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Fig. 2. Characterization of RBSiC. (a) BSE image of RBSiC; (b) Free Si distribution; (c) Porosity distribution; (d) XRD pattern of RBSiC.

Fig. 3(a) presents the BSE image of the brazing alloy surface along with the corresponding EDS analysis. The microstructure primarily consists of white PrSi_2 phase (P1) embedded in the Si matrix (P2), with the overall structure composed of primary PrSi_2 phase and eutectic microstructure ($\text{PrSi}_2 + \text{Si}$). Fig. 3(b) shows the DSC curve, revealing that the melting point of the alloy is 1220°C , which is 8°C higher than the eutectic temperature in the phase diagram, indicating that the alloy composition slightly deviates from the eutectic composition. Fig. 3(c) presents the XRD pattern, further confirming that the primary phases in the filler material are PrSi_2 and Si, with their characteristic diffraction peaks aligning well with the expected results. Additionally, Fig. 3(d) displays the CTE measurement, where the CTE value of the Si-Pr system is determined to be $8.4 \times 10^{-6} \text{ K}^{-1}$ (Fig. 3(d)).



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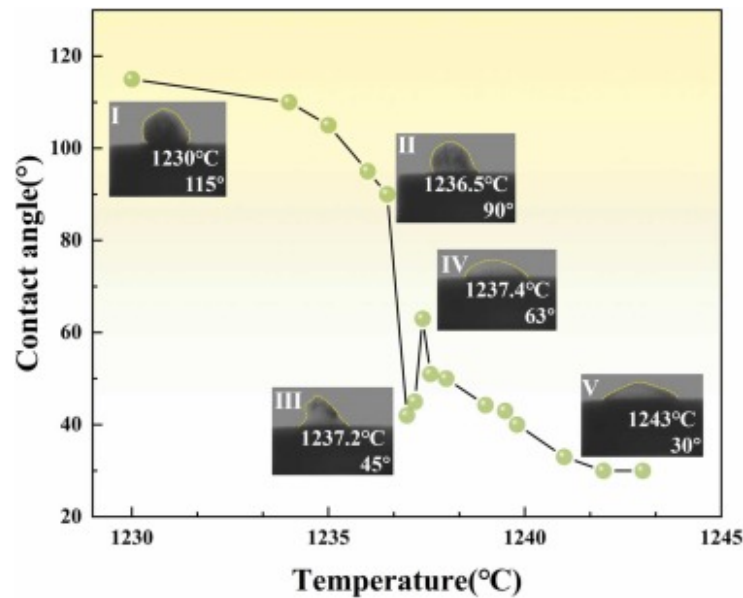
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Fig. 3. Characterization of the Si-Pr eutectic filler. (a) Microstructure (BSE); (b) DSC curve; (c) XRD diffraction pattern; (d) CTE curve.

The formation of the eutectic microstructure is primarily influenced by the solidification dynamics of the eutectic reaction. As the melt cools to the eutectic temperature, Si and PrSi₂ nucleate and grow simultaneously, ultimately forming a typical interwoven eutectic structure (Fig. 3(a)). Both Si and PrSi₂ possess faceted structures, the morphology of the eutectic phases is determined by the competitive growth mechanism between Si and PrSi₂.

Furthermore, local variations in the relative proportions of PrSi₂ and Si phases within the microstructure suggest a possible deviation from the ideal eutectic composition. This deviation may result from solute segregation, localized temperature fluctuations, or non-equilibrium solidification during the solidification process. Consequently, in addition to the eutectic matrix, blocky PrSi₂ phases (P1) were observed in certain regions in the BSE image (Fig. 3(a)), indicating that localized Pr enrichment may have occurred, leading to the preferential precipitation of PrSi₂ at temperatures above the eutectic point (>1212°C). The formation of these blocky PrSi₂ phases is primarily attributed to solute segregation and a relatively slow cooling rate, which allowed sufficient time for PrSi₂ particles to coarsen and grow. This phenomenon is consistent with findings in other metal-silicide eutectic systems, where the growth of faceted phases (such as PrSi₂) is significantly influenced by the temperature gradient and cooling rate [33].

Wetting is a critical factor in brazing, influencing not only the mechanical properties and stability of the joint but also the efficiency and quality of the brazing process. Fig. 4 illustrates the wetting behavior of the Si-Pr filler on RBSiC. The curve illustrates the variation of the apparent contact angle of the molten Si-Pr filler on the RBSiC surface with temperature. The wetting process consists of two key stages: lateral diffusion and vertical diffusion, both of which play pivotal roles in the formation of the wetting. Initially, when the filler contacts the substrate, the presence of a surface oxide film causes the apparent contact angle to decrease gradually. After stage II, the interaction between the filler and the substrate removes the oxide film. The mechanism behind this removal process will be discussed later. As a result, the edge of the droplet rapidly expands outward, leading to a further decrease in the apparent contact angle. After stage III, vertical diffusion occurs. During this stage, the diffusion of elements from the filler into the substrate, along with the enhanced fluidity of the filler at elevated temperatures, leads to a rapid reduction in droplet height, resulting in a more flattened droplet shape [34], [35]. Once equilibrium is reached, the surface tension balances with the internal forces within the liquid, stabilizing both the droplet height and the apparent contact angle.

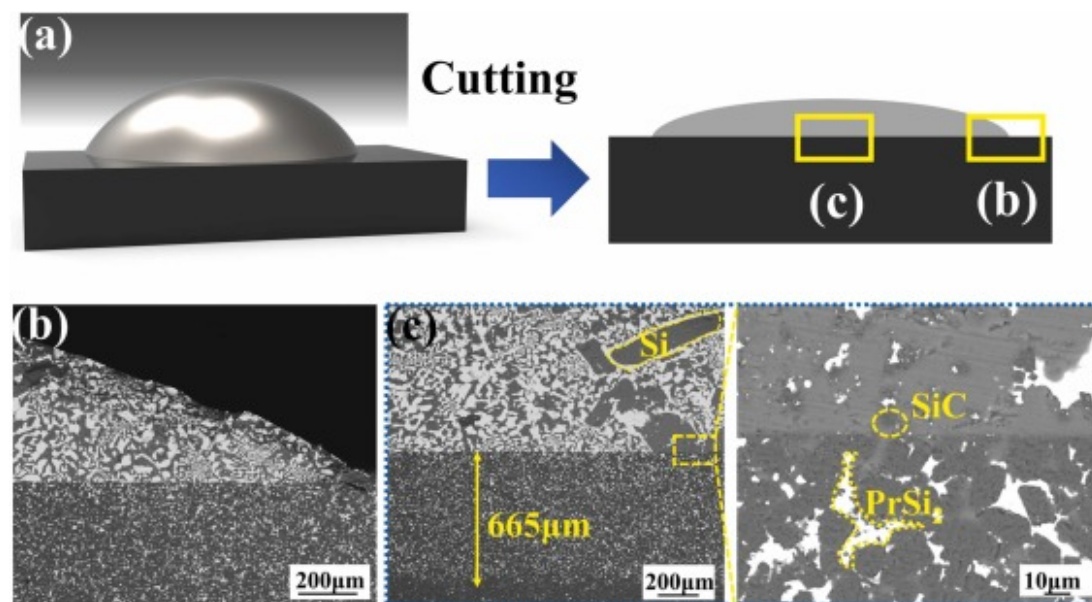


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Fig. 4. The curve of the apparent contact angle of Si-Pr filler on the surface varying with temperature.

Fig. 5(a) presents a schematic diagram of the cross-section of the wetted regions of the Si-Pr filler after 15 min at 1250°C, which exceeds the melting temperature of the filler. Fig. 5(b) shows the microstructural morphology of the wetted contact angle. As the filler spreads laterally, infiltration occurs at the substrate surface, leading to the formation of a diffusion layer with a depth of 665 μm (Fig. 5(c)). In the partially enlarged view of Fig. 5(c), discrete SiC particles are observed near the filler-substrate interface, with SiC primarily surrounding PrSi_2 .



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Fig. 5. Wetting behaviour of Si-Pr filler on RBSiC surface. (a) Schematic diagram of the longitudinal cross-section of the wetted joint; (b)-(c) Microstructure of the wetted joint at the locations marked in (a).

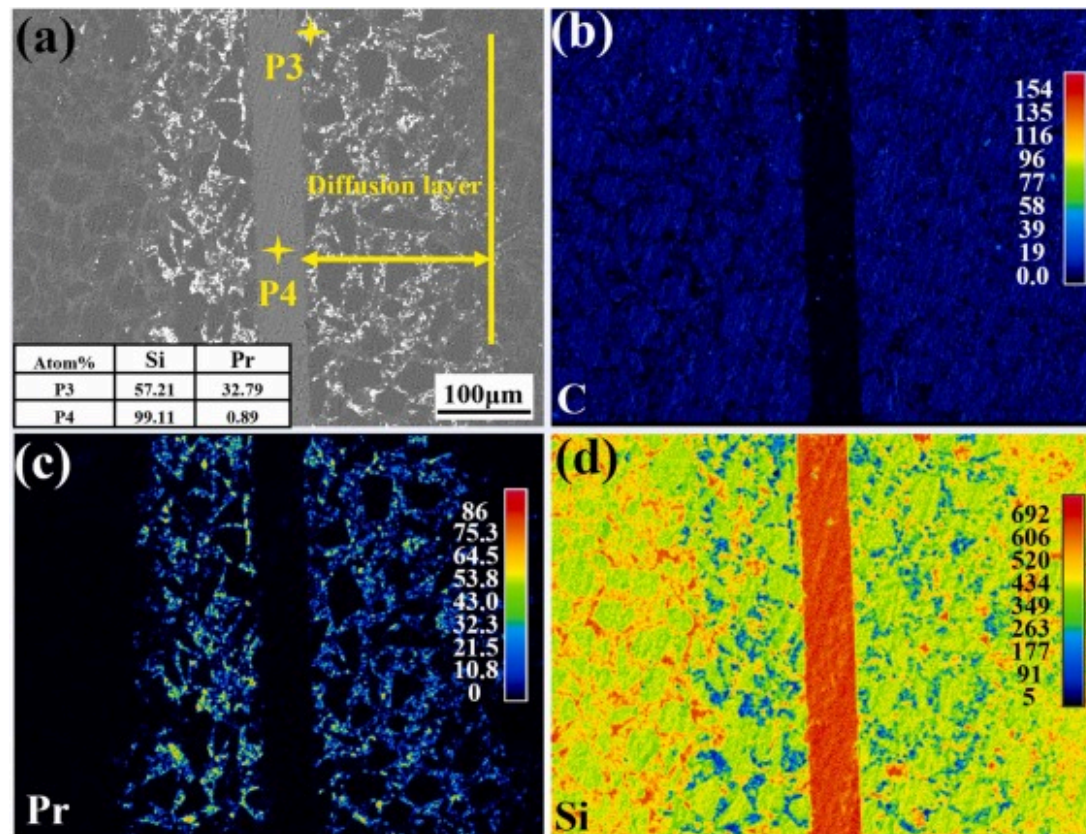
As Pr atom diffuses into the substrate, the free Si within the substrate dissolves, weakening the bonding between free Si and SiC. This dissolution process leads to the detachment of some SiC particles into the alloy, ultimately resulting in the formation of irregularly shaped SiC particles within the microstructure. Below 1100 °C, SiC surface is generally covered by a SiO₂ film, which acts as a wetting barrier [36]. However, when the filler contains Si, it can react with SiO₂ at the interfacial triple-phase line, generating volatile SiO, thereby effectively removing the wetting barrier [37], [38]. Under the experimental conditions of 1250°C and a vacuum level of 2.7×10^{-5} Pa ($\approx 2.7 \times 10^{-10}$ atm), the oxygen partial pressure is sufficiently low to thermodynamically favor the reduction of SiO₂, facilitating the deoxidation process [39]. Previous studies on Si-based brazing fillers have demonstrated that the diffusion kinetics of Si-rich fillers are primarily governed by the rate at which the wetting barrier (SiO₂) is removed from the SiC surface. The following chemical reactions, which are expected to proceed efficiently under these conditions, illustrate this

process [39]:



3.2. Micromorphology and mechanical property of the joint

[Fig. 6](#) shows the microstructure and [EPMA](#) mapping of the joint between Si-Pr filler and RBSiC brazed at 1330 °C for 15 min. The brazed joint is continuous, with no obvious defects. A distinct diffusion layer is formed on both sides of the brazed seam, where the Pr element from the filler diffuses into the substrate in the form of PrSi_2 (P3) compounds. As an active element, Pr facilitates rapid and long-range diffusion, with the diffusion layer exceeding 200 μm in thickness. The brazed seam is primarily composed of Si (P4), which contributes to the joint's excellent high-temperature stability and oxidation resistance. No new phases were observed, further ensuring the stability of the joint. In summary, the stable bonding of the joint is achieved through chemical reactions and element redistribution within the diffusion layer.



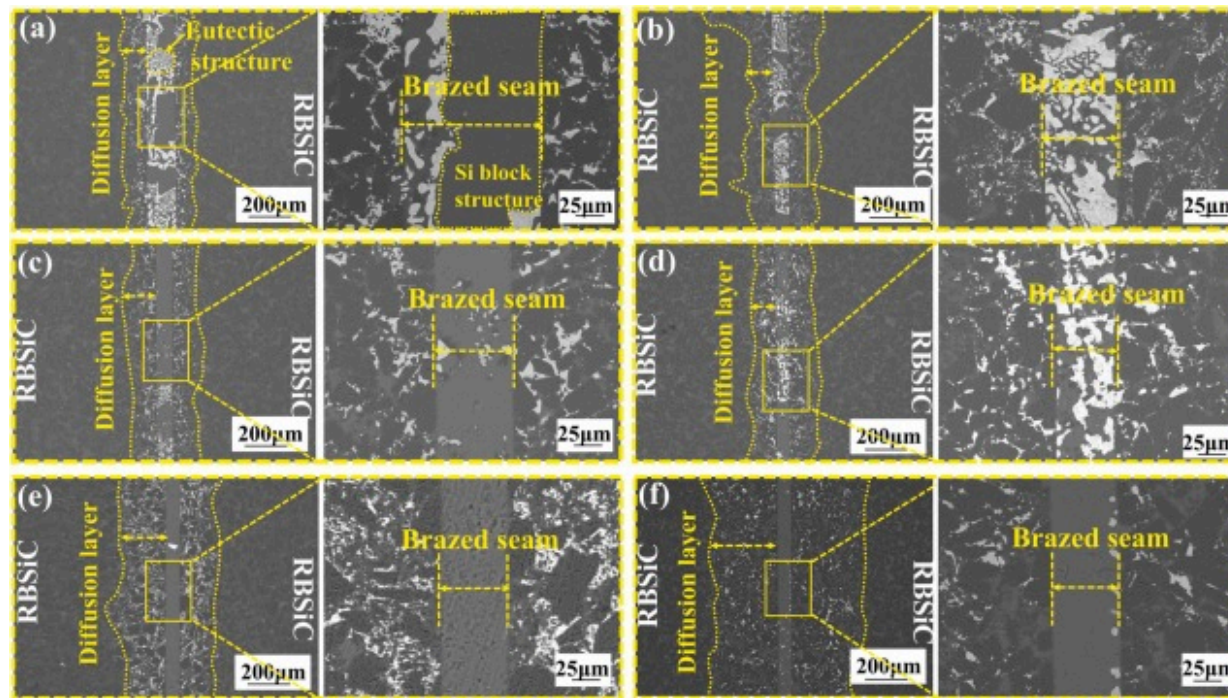
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Fig. 6. EPMA mapping results of the typical joint microstructure of RBSiC at 1330 °C /15 min.

To further investigate the bonding mechanism, brazing experiments were conducted at varying temperature parameters (1250 °C~1350 °C, 15 min) to systematically study the evolution of the joint microstructure. As shown in Fig. 7, with increasing temperature, the brazed seam width significantly narrows, while the thickness of the diffusion layer gradually increases. This narrowing effect is primarily attributed to two factors: (1) the enhanced diffusion of Pr atoms into the substrate at elevated temperatures, which reduces the filler material within the brazed seam, and (2) the formation of PrSi_2 in the diffusion layer, which undergoes a volumetric contraction of approximately 21 %, further contributing to the seam width reduction. This increase in diffusion layer thickness is mainly driven by the temperature effect, where higher temperatures enhance atomic mobility. As

the temperature rises, the thermal activation of Pr atoms increases, allowing them to overcome diffusion energy barriers more efficiently, thereby promoting long-range diffusion into the substrate. Additionally, it accelerates the dissolution dynamics of free Si in the molten eutectic Si-Pr filler, providing a broader diffusion path for the Pr atoms [40].



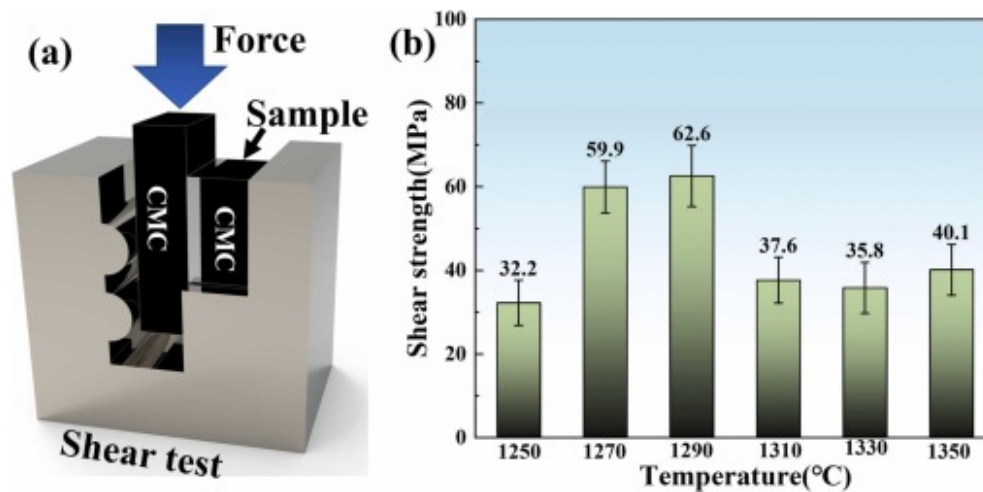
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Fig. 7. Microstructure of joints held at different brazing temperatures for 15 min. (a) 1250 °C; (b) 1270 °C; (c) 1290 °C; (d) 1310 °C; (e) 1330 °C; (f) 1350 °C.

The progressive evolution of the microstructure reveals the non-uniform aggregation of Si phases within the brazed seam. At lower temperatures (Fig. 7(a) and 7(b)), Si appears in blocky structures that progressively grow outward. By 1350 °C (Fig. 7(f)), the brazed seam is predominantly composed of Si, and its microstructure exhibits a highly self-organized Si-rich hypereutectic characteristic. This behavior indicates that the gradient diffusion of Pr atoms drives the filler from a eutectic equilibrium state to a Si-enriched hypereutectic state. Excess Si begins to nucleate and precipitate under the influence of the diffusion gradient, with the precipitates growing dynamically over time.

To evaluate the mechanical properties of the brazed joints, mechanical tests were performed on the joints obtained through different brazing processes, as illustrated in the schematic diagram in Fig. 8(a). Fig. 8(b) presents the shear strength results at different temperatures, showing a trend of increasing and then decreasing strength. At 1250°C, the joints exhibited poor mechanical properties, likely resulting from insufficient bonding between the filler and the substrate. As the temperature increased, the mechanical properties of the joints improved, reaching a peak value of 62.6 MPa at 1290°C. However, at higher temperatures, joint strength decreased, with a slight recovery observed at 1350°C.



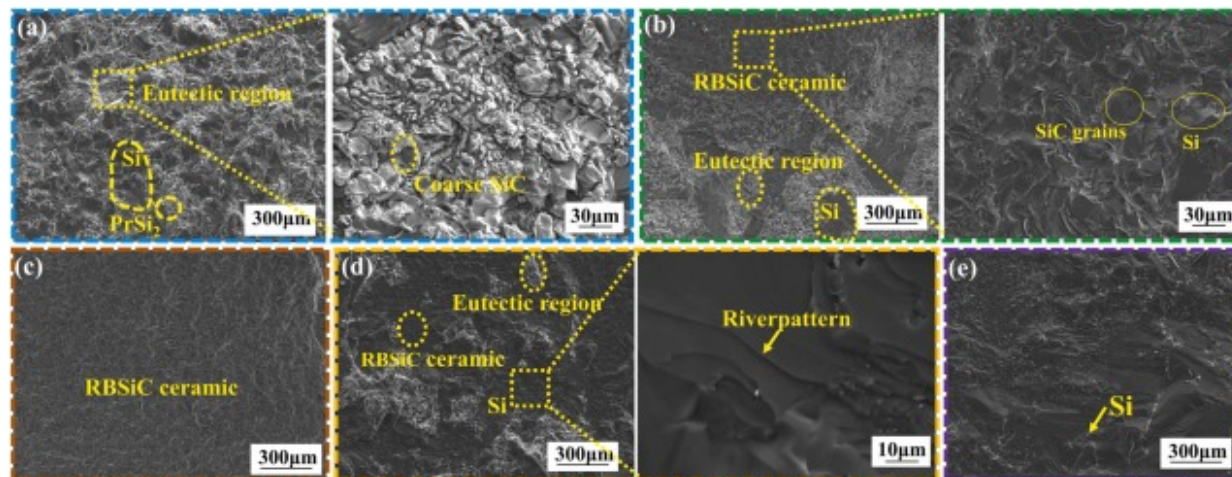
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Fig. 8. Effect of brazing temperature on shear strength of joints. (a) Shear strength test diagram; (b) Shear strength of the joints at different temperatures.

This variation in shear strength is closely related to the microstructural evolution of the joint. At 1290°C, a well-developed diffusion layer ensures strong interfacial bonding, while the moderate formation of PrSi₂ contributes to enhanced joint strength without introducing excessive brittleness. As the temperature increases further, the intensified diffusion of Pr into the substrate depletes Pr in the brazed seam, leading to the formation of a Si-rich hypereutectic structure, which reduces joint ductility and consequently lowers shear strength. The observed variations in shear strength can be further analyzed through fracture surface morphology examination.

The fracture morphology of the joints at different brazing temperatures is shown in Fig. 9. At 1250 °C, the fracture surface is entirely covered with filler (Fig. 9(a)), indicating that the joint fractured at the brazed seam. The magnified view reveals prominent SiC grains and PrSi₂ protrusions, which are considered critical microstructural features for enhancing the mechanical performance of the joint. However, due to insufficient melting and diffusion of the filler, as well as the inherent brittleness of Si-based fillers, the macroscopic strength of the joint remains limited. At 1270 °C, the fracture surface exhibits a coexistence of SiC and Si-Pr filler, accompanied by the formation of blocky Si within the brazed seam (Fig. 9(b)). At this stage, the fracture mode transitions to a mixed fracture involving both the brazed seam and the substrate, indicating an enhancement in interfacial reactions between the filler and the substrate. As the temperature increases to 1290 °C, fracture occurs almost entirely within the ceramic substrate (Fig. 9(c)), indicating that the bonding strength of the joint reaches its peak at this temperature. The significant shift in fracture mode corresponds to the maximum shear strength, corroborating the temperature-dependent regulation of the joint's mechanical behavior. This is primarily due to the diffusion of Pr atoms into the substrate and the formation of a stable PrSi₂ phase at the interface, as evidenced by the SEM images (Fig. 7). The PrSi₂-rich region at the brazed seam-substrate interface plays a crucial role in enhancing the interface strength. Studies have shown that the infiltration of metal silicides can significantly improve the hardness of SiC composites, thereby contributing to the overall mechanical stability of the joint [41], [42]. At 1310 °C, the fracture surface becomes dominated by large areas of Si (Fig. 9(d)), and the magnified view reveals typical river-like brittle fracture patterns of Si, indicating deep diffusion of Pr atoms into the substrate. This microstructural feature indicates that the bonding mechanism shifts from interfacial chemical reactions to being primarily governed by adsorption of Si on SiC. However, this transition is accompanied by an increased propensity for brittle fracture, leading to a deterioration in mechanical performance. At 1330 °C, the fracture surface is almost entirely composed of Si (Fig. 9(d)), where the formation of a highly Si-enriched phase at elevated temperatures significantly weakens the mechanical stability of the joint, resulting in a pronounced decline in strength.



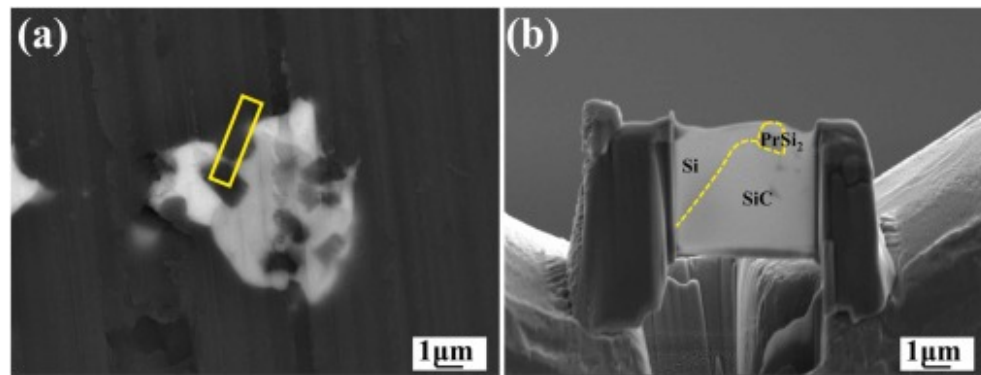
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Fig. 9. Fracture morphology at different brazing temperatures. (a) 1250 °C; (b) 1270 °C; (c) 1290 °C; (d) 1310 °C; (e) 1330 °C.

3.3. Evolutionary mechanisms of brazed joint

Fig. 10(a) shows the FIB sampling location at the brazed seam, containing Si, PrSi₂, and SiC phases. The sample was extracted along the region marked by the yellow box and further processed through precise longitudinal cutting, resulting in an FIB sample with final dimensions of 5 µm × 5 µm. After cutting, the distinct grain structures within the material became clearly visible (Fig. 10(b)), effectively revealing its microstructural characteristics and providing a reliable structural foundation for subsequent high-resolution TEM analysis.

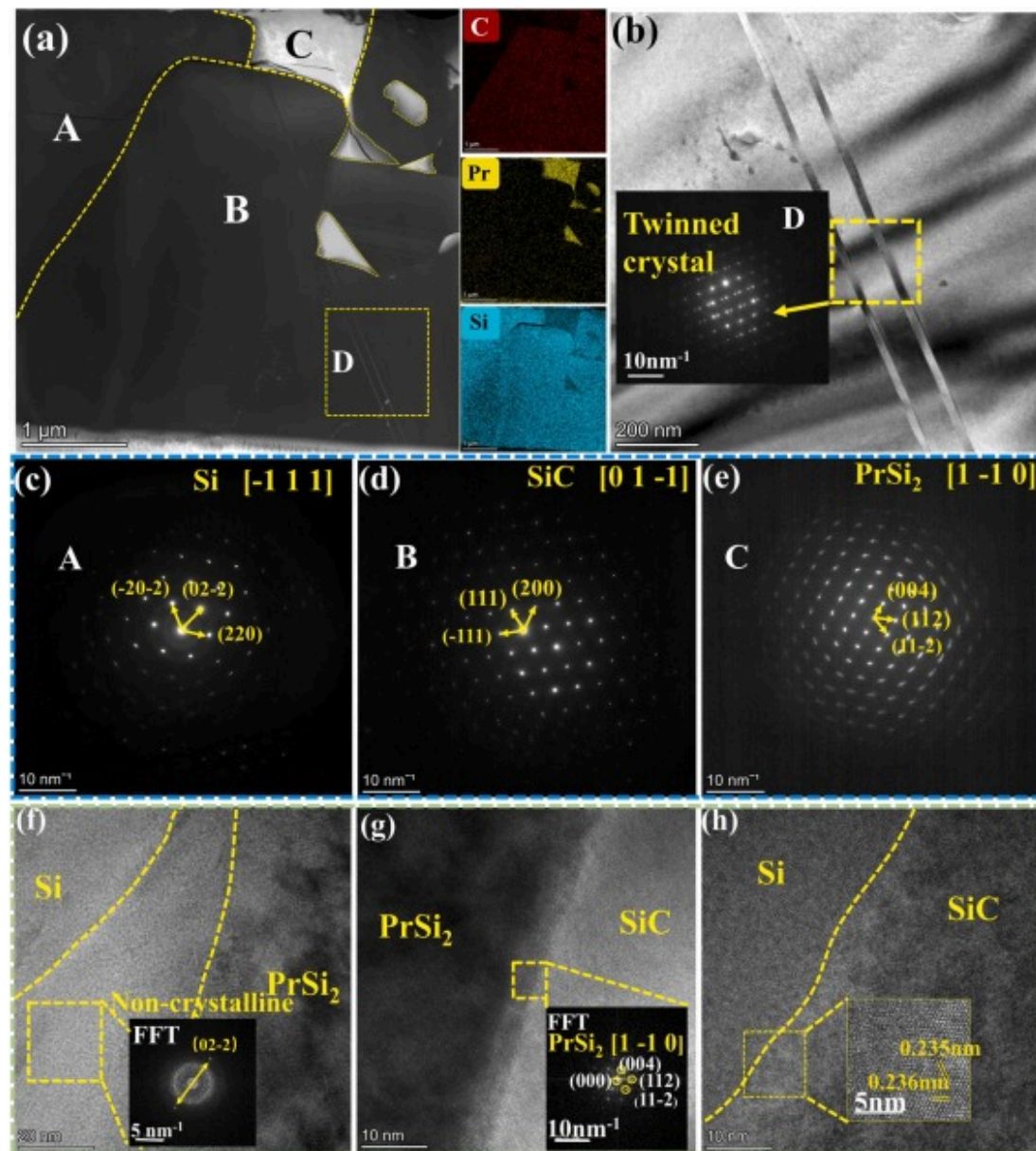


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Fig. 10. FIB-manufactured sample. (a) Microstructure at the sampling location; (b) Microstructure of the TEM sample.

Fig. 11 presents the TEM images and elemental distribution maps of the samples prepared by FIB. According to SAED analysis (Fig. 11(c-e)), the brazed seam consists of cubic-structured Si, cubic-structured SiC, and tetragonal-structured PrSi_2 . Notably, the Si-Pr filler itself does not contain SiC phases. Based on the microstructural analysis of Fig. 5(c), the diffusion of Pr atoms into the substrate may weaken the bonding between free Si and SiC. This weakening could lead to the detachment of some SiC particles into the brazed region, ultimately forming irregularly shaped SiC within the joint. Further SAED analysis reveals the presence of twin defects in the SiC grains that have detached from the substrate (Fig. 11(b)), likely due to thermal expansion mismatch and interfacial stresses.



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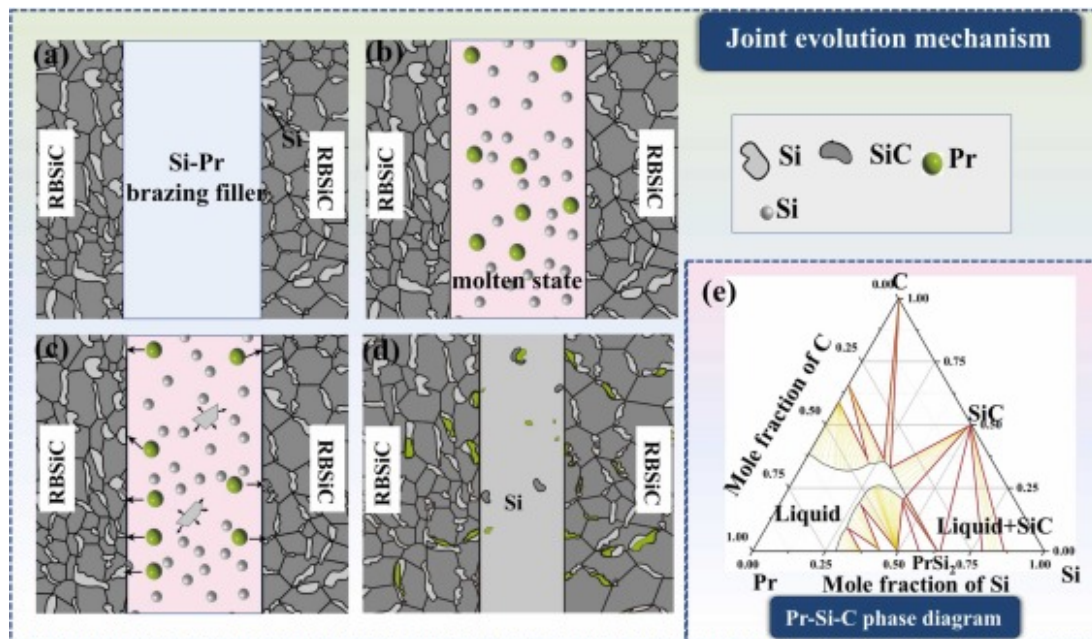
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Fig. 11. TEM image of the brazed region. (a) HADDF and elemental distribution map; (b) BF image and SEAD analysis of region D in (a); (c)-(e) SEAD analysis of region A-C in (a); (f)-(h) HRTEM images and corresponding FFT analysis of various phase boundaries.

HRTEM images of the interface reveal excellent bonding quality, with a one-dimensional disordered amorphous Si structure observed between PrSi₂ and Si (Fig. 11(f)). This phenomenon is attributed to the significant lattice mismatch between the two phases, resulting in interatomic stresses that cannot be relieved through conventional elastic deformation, thereby leading to the formation of amorphous or semi-amorphous regions at the interface. Fourier transform analysis of the PrSi₂/SiC interface region (Fig. 11(g)) confirms that no new phases are formed during the diffusion bonding process, and no intermediate bonding phases are present at the PrSi₂ interface. The bonding at this interface is primarily governed by the long-term diffusion of the Si-Pr filler, resulting in the development of atomic bonding forces at the interface. Similarly, the Si and SiC interface also exhibits strong bonding with no intermediate phase formation (Fig. 11(h)).

In summary, the bonding mechanism does not rely on the formation of intermediate compounds but rather on the chemical adsorption of Si atoms on SiC, as well as the atomic bonding forces arising from the melting of the Si-Pr filler and the subsequent reconstruction at the interface. This bonding mechanism highlights the critical role of diffusion in stabilizing the interface and ensuring the mechanical performance of the joint during high-temperature brazing.

Based on the previous analysis, the reaction mechanism for joining Si-Pr filler with RBSiC is illustrated in Fig. 12. At the eutectic temperature, the Si-Pr filler melts and undergoes interfacial reactions with the substrate (Fig. 12(b)). During the brazing process, the Si-rich filler performs two critical functions. First, it inhibits the direct reaction between activated Pr and SiC, preventing the formation of graphite phases. Second, it facilitates the solid-phase dissolution of Si from the substrate, thereby establishing a Pr concentration gradient between the brazed seam and the molten Si. Simultaneously, the dissolution of Si further induces a Pr concentration gradient within the substrate, driving the diffusion of Pr along the Si distribution pathway (Fig. 12(c)). As the diffusion process progresses, the system gradually approaches thermodynamic equilibrium, leading to the precipitation of Pr in the brazed seam as PrSi₂ (Fig. 12(d)).



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Fig. 12. Diagram of the microstructure evolution mechanism of the joining Si-Pr filler with RBSiC. (a) Initial state of the interface; (b) Melting of the filler metal; (c) Expansion reaction stage; (d) Cooling stage; (e) Pr-Si-C phase diagram at 1300°C [43].

Although Pr atoms do not directly react with SiC, the interaction between the brazing filler and the SiC interface alters the local thermodynamic equilibrium, as indicated by the Pr-Si-C ternary phase diagram, thereby inducing the mechanical detachment of SiC particles. As Si dissolves from the substrate into the molten filler, the local composition evolves along the liquid-phase region of the ternary phase diagram, shifting towards the liquid + SiC equilibrium region. This shift weakens the interfacial bonding forces of SiC, ultimately leading to its detachment. The detached SiC particles disperse within the molten filler and, upon solidification, become embedded in the brazed seam, thereby enhancing the thermal expansion coefficient compatibility of the joint and providing structural reinforcement to optimize its mechanical properties.

As the Pr concentration in the brazed seam decreases, the composition of the molten filler deviates from the eutectic point, inducing the nucleation and growth of Si precipitates. This precipitation process further increases the concentration of Pr in the

liquid phase, sustaining the diffusion driving force. The continuation of this diffusion process results in the formation of a deeper diffusion layer at the substrate interface, laying the microscopic foundation for the stable bonding and enhanced performance of the joint.

When Pr elements come into contact with SiC, SiC may decompose, leading to graphite precipitation, which can adversely affect the stability of the joint. However, Si effectively inhibits this reaction, preventing the formation of graphite at the interface. This inhibitory mechanism can be quantitatively analyzed through thermodynamic calculations. At 1250°C, the standard Gibbs free energy for the formation of SiC from graphite and liquid Si is -56.389 J/mol (Eq. (3)). Based on the regular solution model, the excess Gibbs free energy of Si in the Pr-Si alloy is expressed by Eq. (4), with $\lambda = -225$ kJ/mol. The calculated equilibrium Si mole fraction ($x_{Si}^* \approx 0.55$) is significantly lower than the actual Si content in the filler (83 %) [44], confirming that the Si-Pr alloy effectively suppresses graphite precipitation and stabilizes SiC. To validate this theoretical prediction, XRD and TEM analyses were conducted, and no graphite phases were detected at the interface, further confirming the stability of SiC in the Si-Pr filler environment. By combining thermodynamic calculations with experimental validation, we confirm that the Si-Pr filler exhibits excellent compatibility with SiC and ensures a stable interfacial bonding mechanism.

$$SiC_{(s)} \rightleftharpoons (Si) + C_{gr} \quad (3)$$

$$\overline{\Delta G}_{Si(Pr)}^{XS} = \lambda_{Si-Pr} (1 - x_{Si})^2 \quad (4)$$

$$\Delta G_{f,SiC}^0 = \overline{\Delta G}_{Si}^{XS} + RT \ln x_{Si}^* \quad (5)$$

4. Conclusion

The brazing of Si-Pr eutectic filler with RBSiC was systematically explored, focusing on the microstructural evolution, mechanical properties, and evolutionary mechanisms of the joints. These findings offer critical insights for the advanced fabrication of large, intricate composite structures subjected to extreme thermal environments, such as those encountered in aerospace and nuclear engineering. The following discussion can be drawn:

- (1) The Si-Pr eutectic filler can rapidly spread on RBSiC at 1250°C. The removal of surface oxides reduces interfacial tension, while the longitudinal diffusion of Pr atoms enhances the filler-substrate interaction, further improving the wetting behavior.

- (2) At 1290°C, the joint shear strength reaches 62.6 MPa, indicating that an appropriate amount of PrSi₂ phase at the interface enhances the joint strength. However, as the temperature further increases, excessive diffusion of Pr into the substrate leads to a reduction in Pr content in the brazed seam, forming a Si-rich hypereutectic structure, which consequently decreases the ductility and shear strength of the joint.
- (3) The bonding between the Si-Pr filler and RBSiC is primarily driven by filler diffusion and interface reconstruction of Si and Pr. No new phase formation was observed; however, interfacial processes such as Si dissolution, Pr diffusion, and SiC detachment contributed to joint stability and performance enhancement.

CRediT authorship contribution statement

Fu Mengchun: Software. **Lin Panpan:** Validation. **Zhang Qianrun:** Validation. **Wu Tong:** Visualization. **Liu Zhanguo:** Methodology, Investigation. **Lin Tiesong:** Project administration, Data curation. **Wang Ce:** Validation. **Dong Yongkang:** Investigation. **Yang Jia:** Software. **Fu Yibo:** Validation. **Xue Xiaoqian:** Writing – original draft, Visualization. **Lu Fugang:** Supervision, Resources. **He Peng:** Funding acquisition, Formal analysis.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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